Complex Analysis - Quick Notes

Lucas Z. Brito

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1 Exponents and Logarithms

- For a polar complex number
$$z = re^{i\theta} \implies |z| = r, \quad \operatorname{Arg} z = \theta$$
- **Principal valued** functions are capitalized like the above always take on values in the interval $(-\pi, \pi]$. This is called the principal branch.
- Corresponding functions in lowercase are defined up to a 2π rotation in the complex plane:
$$\arg z \equiv \{\operatorname{Arg} z + 2\pi k \mid k \in \mathbb{Z}\}$$
- The **exponential function** is *defined* by its usual Taylor series. Note that
$$e^{z_1+z_2} = e^{z_1}e^{z_2}$$
$$e^{z_1-z_2} = e^{z_1}/e^{z_2}$$
$$(e^z)^n = e^{nz}, \quad n \in \mathbb{Z}$$
but it is *not* generally true that
$$(e^{z_1})^{z_2} = e^{z_1 z_2}$$
- **Complex logarithm** defined by
$$\log z = \log|z| + i\arg z, \quad \operatorname{Log} z = \log|z| + i\operatorname{Arg} z$$
The first formula gives all complex numbers $w = \log z$ such that $e^w = z$. The second gives the single-valued inverse of e^w for $\operatorname{Im} w \in (-\pi, \pi]$.

  You can imagine this definition is motivated by
$$\log re^{i\theta} = \log r + \log e^{i\theta} = \log|z| + i\theta$$
except θ is ambiguous up to $2\pi k$, $k \in \mathbb{Z}$, hence the above definition.

- The complex **power function** is defined by

$$z^\alpha \equiv e^{\alpha \log z} = e^{\alpha \operatorname{Log} z} e^{2\pi i \alpha k}, \quad k \in \mathbb{Z}$$

- Trigonometric functions:

$$\cos z = \frac{1}{2}\left(e^{iz} + e^{-iz}\right), \quad \sin z = \frac{1}{2i}\left(e^{iz} - e^{-iz}\right)$$

2 Holomorphic Functions

- **Holomorphic**: complex differentiable. Limit needs to approach the same value regardless of the direction you take. Complex differentiability can be defined by two formulas:

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0), \quad f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0)$$

- **Cauchy-Riemann equations**: roughly state that it doesn't matter the direction from which you approach a value, the limit is still the same.

$$u_x = v_y, \quad u_y = -v_x$$

  or in polar form:

$$u_r = \frac{v_\theta}{r}, \quad v_r = -\frac{u_\theta}{r}$$

  Note that obeying the Cauchy-Riemann equations is synonymous with being holomorphic.

- If $f = u + iv$ is holomorphic and u, v have continuous second order derivatives, u and v are harmonic.
- **Analytic continuation:** $B \subset A$, A, B domains. Then analytic continuation $\tilde{f}(z)$ of $f(z)$ is
  1. $\tilde{f}(z) = f(z) \quad \forall z \in B$.
  2. $\tilde{f}(z)$ analytic on A.
- **Entire function** is analytic in all of $\mathbb{C}$.

3 Complex Integration

- **Green's Theorem**:

$$\int_{\partial D} P \,\mathrm{d}x + Q \,\mathrm{d}y = \iint_D \mathrm{d}x \,\mathrm{d}y \, Q_x - P_y$$

- Differentials:
  - $P \,\mathrm{d}x + Q \,\mathrm{d}y$ is **exact** if $P \,\mathrm{d}x + Q \,\mathrm{d}y = \mathrm{d}h$
  - Differential **closed** if $P_y = Q_x$.
  - Exact differentials are closed.
  - If on a star-shaped domain, closed differentials are exact in that domain.
- The **complex line integral** is defined by

$$\int_\gamma h(z) \,\mathrm{d}z = \int_\gamma h(z) \,\mathrm{d}x + i \int_\gamma h(z) \,\mathrm{d}y$$

- **Estimation lemma**:

$$\left| \int_\gamma h(z) \,\mathrm{d}z \right| \leq \int_\gamma |h(z)| |\,\mathrm{d}z\,| \leq L(\gamma) \cdot \sup_{z \in \gamma} |h(z)|, \quad L(\gamma) \equiv \text{length of } \gamma$$

- **Cauchy Integral Formula**:

$$f(z) = \frac{1}{2\pi i}\int_{\partial D} \frac{f(w)}{w-z}\,\mathrm{d}w\,, \quad f^{(m)}(z) = \frac{m!}{2\pi i}\int_{\partial D} \frac{f(w)}{(w-z)^{m+1}}\,\mathrm{d}w$$

  Note: when the function two singularities of the form $(w-z)^n$ in the denominator, don't forget to, for different line integrals, incorporate the singularities not contained by the path in $f(z)$. Also note that if your path contains two singularities, you can just split it into two paths surrounding each singularity. See here for pole *on* contour.

- **Liouville's Theorem**: $f(z)$ holomorphic on complex plane; if $f(z)$ bounded, it is constant.

- **Morera's Theorem**: $f(z)$ continuous function on a domain D; If $\int_{\partial R} f(z)\,\mathrm{d}z = 0$ for all closed rectangles R contained in D with sides parallel to coordinate axes, $f(z)$ is analytic on D.

- **Goursat's Theorem**: If $f(z)$ on domain D such that

$$f'(z_0) = \lim_{z\to z_0} \frac{f(z)-f(z_0)}{z-z_0}$$

  exists at each $z_0 \in D$, $f(z)$ analytic on D.

4 Power Series

- **Weierstrauss M-Test:** $M_k > 0$ and $\sum M_k$ converges; $g_k(x) : E \to \mathbb{C}$ for some set E s.t. $|g_k(x)| \leq M_k$ for all $x \in E$. Then $\sum g_k(x)$ converges uniformly on E.

- **Power series:** a series of the form

$$\sum_{k=0}^{\infty} a_k (z-z_0)^k$$

  Can make change of variables to translate to

$$\sum_{k=0}^{\infty} a_k z^k$$

  for which there is $R \in [0, +\infty]$ s.t., the sum converges absolutely if $|z| < R$ and the sum does not converge if $|z| > R$. This function is analytic for $|z| < R$.

- **Ratio test:** If $|a_k/a_{k+1}|$ has a limit as $k \to \infty$, either finite or $+\infty$, then the limit is the radius of convergence of $\sum a_k z^k$.

- **Power series expansion at infinity:** $f(z)$ analytic at ∞ if $g(w) = f(1/w)$ analytic at $w = 0$. *Generally this change of variables will constitute definitions at infinity.* Power series expansions would be

$$g(w) = \sum_{k=0}^{\infty} b_k w^k, \quad |w| < \rho \quad \Longleftrightarrow \quad f(z) = \sum_{k=0}^{\infty} \frac{b_k}{z^k}, \quad |z| > \frac{1}{\rho}$$

- **Zeros:** zero of order N at z_0 if $f(z_0) = f'(z_0) = \cdots = f^{(N-1)}(z_0) = 0$ (derivative of order N is nonzero). Holds iff

$$f(z) = a_N(z-z_0)^N + a_{N+1}(z-z_0)^{N+1} + \cdots \Longrightarrow f(z) = (z-z_0)^N h(z)$$

  Zero at $z = \infty$ if $g(w) = f(1/w)$ has zero at $w = 0$. Zeros of analytic function are isolated (if the function is not identically zero).

5 Laurent Series

- **Laurent decomposition:** $0 \leq \rho < \sigma \leq \infty$. $f(z)$ analytic for annulus $\rho < |z - z_0| < \sigma$. Then $f(z)$ can be written

$$f(z) = f_0(z) + f_1(z), \qquad \begin{cases} f_0(z) \text{ analytic for } |z - z_0| < \sigma \\ f_1(z) \text{ analytic for } \rho < |z - z_0| \end{cases}$$

  Unique if enforce $f_1(\infty) = 0$. That is, function can be split into part analytic within the outer bound of annulus and a part analytic outside the inner bound of the annulus.

- **Laurent series:** Power series expansion but with negative powers.

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k, \qquad \rho < |z - z_0| < \sigma$$

  - Annulus of convergence's upper and lower radii extend up until it hits a pole. So the size of the annulus of convergence is determined the poles of the function.
  - Given an annulus, the Laurent series for a given function is unique.
  - The coefficients are given by either manipulating the series or

$$a_k = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}}$$

    where C is a closed curve *contained* in the annulus.
  - Converges absolutely at each point of the annulus; converges uniformly on each subannulus $\rho < r \leq |z - z_0| \leq s < \sigma$.

Example 5.1: Laurent series

Taken from Wikipedia. Take this function and its partial fraction decomposition:

$$f(z) = \frac{1}{(z-1)(z-2i)} = \frac{1+2i}{5}\left(\frac{1}{z-1} - \frac{1}{z-2i}\right) \qquad \text{singularities at } 1, 2i$$

Three possible Laurent expansions converging in different regions (centered at zero):

- On the disk $|z| < 1$. The same as the Taylor series. We can obtain the coefficients by integrating along a circle of radius less than 1.
- Middle annulus $1 < |z| < 2$, caught between the two singularities. We can obtain the coefficients by integrating along a circle of radius less than 2 but greater than 1.
- On the (infinite) outer annulus $2 < |z| < \infty$. Also the Laurent series at infinity. We can obtain the coefficients by integrating along a circle of radius greater than 2.

- **Isolated singularities:** z_0 is an isolated singularity if $f(z)$ analytic in the punctured disk $0 < |z - z_0| < r$. E.g., $\sqrt{z}$ and $\log z$ do not have isolated singularities at $z = 0$. They are singular along some branch cut. Laurent series has expansion

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z_k - z_0)^k \qquad 0 < |z - z_0| < r \tag{1}$$

  With respect to the above series:

  - **Removable singularity** if $a_k = 0$ for all $k < 0$. E.g., $\sin x / x$.

- **Pole** if $N > 0$ such that $a_{-N} \neq 0$ but $a_k = 0\ \forall k < -N$. N is the order of the pole. $N = 1$: simple pole, $N = 2$ double pole.
- **Essential singularity** if $a_k \neq 0$ for infinitely many $k < 0$.

Note we can "flip" the function: z_0 an isolated singularity of $f(z)$. Then z_0 is a pole of order $N \iff 1/f(z)$ analytic at z_0 and has a zero of order N at z_0

Isolated singularity at ∞ if $f(z)$ analytic there is $R > 0$ such that $f(z)$ is analytic for $|z| > R \iff g(w) = f(1/w)$ has isolated singularity at $w = 0$. Classification of singularities at infinity are natural based on classification of singularity at 0 of $g(w)$.

- **Riemann's theorem on Removable Singularities.** z_0 isolated singularity. Then if $f(z)$ bounded near z_0, it is a removable singularity.
- **Meromorphic function:** a function is meromorphic on D domain if $f(z)$ is analytic on D except at isolated singularities, each of which is a pole.
- *Debunking* $1 + 2 + 3 + \cdots = -1/12$. Zeta function has value $\zeta(s = -1) = -1/12$, and *for the values* $\mathrm{Re}(s) > 1$, is given by series

$$\zeta(s) \sum_{n=1}^{\infty} \frac{1}{n^s}$$

such that setting $s = -1$ recovers $1 + 2 + 3 + \cdots$. But outside $\mathrm{Re}(s) > 1$ it is given by its analytic continuation, *not the sum.* So we can't say the above sum with $s = -1$ is equal to $\zeta(-1)$.

6 Residue Calculus

- **Residue:** z_0 is an isolated singularity. Laurent series given by eq. (1). Then

$$\mathrm{Res}[f(z), z_0] = a_{-1} = \frac{1}{2\pi i} \oint_{|z-z_0|=r} f(z)\,\mathrm{d}z$$

- **Residue theorem:** $f(z)$ analytic on domain D and its boundary, except isolated singularities $z_1, z_2, ...z_m$. Then

$$\int_{\partial D} f(z)\,\mathrm{d}z = 2\pi i \sum_{j=1}^{m} \mathrm{Res}[f(z), z_j]$$

  - Simple pole at z_0: $\mathrm{Res}[f(z), z_0] = \lim_{z\to z_0}(z - z_0)f(z)$.
  - Double pole at z_0: $\mathrm{Res}[f(z), z_0] = \lim_{z\to z_0} \frac{d}{dz}\left[(z - z_0)^2 f(z)\right]$.
  - $f(z)$ and $g(z)$ are analytic at z_0, $g(z)$ has simple zero at z_0: $\mathrm{Res}[f(z)/g(z), z_0] = f(z_0)/g'(z_0)$.
  - $g(z)$ analytic and has simple zero at z_0: $\mathrm{Res}[1/g(z), z_0] = 1/g'(z_0)$.
- **Integrals of Rational Functions:** general approach for real integrals of rational functions is to turn into a contour integral: e.g. integral over $(-\infty, \infty)$

$$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x$$

into large semicircular contour with arc Γ_R and line from $-R$ to R, then let $R \to \infty$, around the pole:

$$\int_{\partial D_R} f(z)\,\mathrm{d}z = \int_{\Gamma_R} f(z)\,\mathrm{d}z + \int_{-R}^{R} f(z)\,\mathrm{d}z$$

LHS can be evaluated with residue theorem, will be some residues. Then use estimation lemma for first term RHS (see section 3) to find something like

$$\mathrm{Res}[f(z)] = O(R^{-n}) + \int_{-R}^{R} f(z)\,\mathrm{d}z \xrightarrow{R\to\infty} \mathrm{Res}[f(z)] = \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x \qquad (\text{if } n > 0)$$

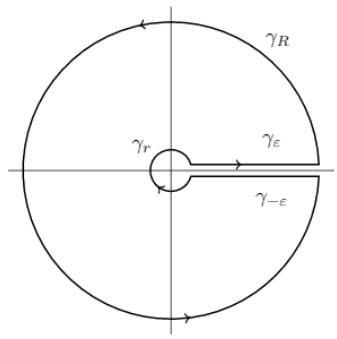

Figure 1: **NOTE** the small ϵ-circle we need to take care of around the origin. Cut the paths along the positive axis off.

- **Integrals of Trigonometric Functions:** integral of form

$$\int_0^{2\pi} R(\cos\theta, \sin\theta)\,\mathrm{d}\theta$$

use $\cos\theta + (e^{i\theta} + e^{-\theta})/2$ and $\sin\theta = (e^{i\theta} - e^{-\theta})/2i$. Turns into

$$\begin{aligned}\int_0^{2\pi} R\left(\frac{e^{i\theta}+e^{-i\theta}}{2}, \frac{e^{i\theta}+e^{-i\theta}}{2i}\right)\mathrm{d}\theta &= -i\int_0^{2\pi} R\left(\frac{\gamma(\theta)+1/\gamma(\theta)}{2}, \frac{\gamma(\theta)-1/\gamma(\theta)}{2i}\right)\frac{\dot{\gamma(\theta)}}{\gamma(\theta)}\,\mathrm{d}\theta \qquad \gamma(\theta) = e^{i\theta} \\ &= -i\int_0^{2\pi} R\left(\frac{z+1/z}{2}, \frac{z-1/z}{2i}\right)\frac{\mathrm{d}z}{z} \\ &= 2\pi\sum_{a\in D}\mathrm{Res}[f(z), a] \qquad f(z) = \frac{1}{z}R\left(\frac{z+1/z}{2}, \frac{z-1/z}{2i}\right)\end{aligned}$$

- **Integrands with Branch Points:** need to make a branch cut; i.e., specify the branch you're using. E.g, specify values for $0 < \theta < 2\pi$. Then integrate along "keyhole contour", see fig. 1. Values on bottom edge of real positive axis differ from each other by some phase factor $f(e^{2\pi i})$ (I think?)

- **Fractional Residues:** if z_0 is a simple pole, C_ϵ an arc of the circle $|z - z_0| = \epsilon$ with angle α,

$$\lim_{\epsilon\to 0}\int_{C_\epsilon} f(z)\,\mathrm{d}z = \alpha i\,\mathrm{Res}[f(z), z_0]$$

- **Residue at infinity:** Use Laurent series that converges for *infinite annulus* $|z| \geq R$. Residue at infinity is negative of -1-th coeff.:

$$\mathrm{Res}[f(z), \infty] = -a_{-1} = \int_{\partial D_R} f(z)\,\mathrm{d}z = -\,\mathrm{Res}\left[\frac{1}{z^2}f\left(\frac{1}{z}\right), 0\right]$$

  - Negative because it's like we're doing an integral enclosing infinity except in the opposite direction because infinity outside of the circle.
  - Note residue associated with differential $f(z)\,\mathrm{d}z$, not function $f(z)$, so that we don't need to worry about our choice of coordinates. Explains $1/z^2$ factor in above definition, for if we make the change of variables $z \to 1/z$:

$$f(z)\,\mathrm{d}z \to -f(z)\frac{\mathrm{d}z}{z^2} \qquad \mathrm{d}\left(\frac{1}{z}\right) = -\frac{\mathrm{d}z}{z^2}$$

  - Also means that, for residue at infinity, some functions can be analytic at infinity while having a residue there.

7 Logarithmic Integral

- **Logarithmic integral:** (assume $f(z) \neq 0$ on γ):

$$\frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z)}\,\mathrm{d}z = \frac{1}{2\pi i}\int_\gamma \mathrm{d}\log f(z)$$

- **Counting zeros and poles:** D bounded domain, piecewise smooth boundary. $f(z) \neq 0$ on ∂D, analytic on boundary.

$$\frac{1}{2\pi i}\int_{\partial D} \frac{f'(z)}{f(z)}\,\mathrm{d}z = (\#\text{ zeros}) - (\#\text{ poles})$$

  where we *count the orders* of the zeros and poles.

- **Increase in argument of function:**

$$\begin{aligned}
\int_\gamma \mathrm{d}\arg f(z) &= \arg f(\gamma(b)) - \arg f(\gamma(a)) \\
-i\int_\gamma \mathrm{d}\log f(z) &= -i\int_\gamma \mathrm{d}\log|f(z)| + \int_\gamma \mathrm{d}\arg f(z) \\
\int_\gamma \mathrm{d}\log|f(z)| &= \log|f(\gamma(b))| - \log|f(\gamma(a))| \\
\int_{\partial D} \mathrm{d}\arg f(z) &= 2\pi(\#\text{ zeros} - \#\text{ poles})
\end{aligned}$$

- **Winding numbers:** $W(\gamma, z_0)$ of γ around z_0.

$$W(\gamma, z_0) = \frac{1}{2\pi i}\int_\gamma \frac{\mathrm{d}z}{z - z_0} = \frac{1}{2\pi}\int_\gamma \mathrm{d}\arg(z - z_0)$$

  D a domain and $f(z)$ analytic in D. γ a closed path in D. $W(\gamma, \zeta) = 0$ for all ζ outside D. Then

$$\frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z - z_0}\,\mathrm{d}z = W(\gamma, z_0)f(z_0)$$

- **Rouché's theorem:** D bounded domain with piecewise smooth boundary. $f(z)$, $h(z)$ analytic on $D \cup \partial D$. $|h(z)| < |f(z)|$ for $z \in \partial D$, then $f(z)$ and $f(z)+h(z)$ have the same number of zeroes in D, counting multiplicities.

Example 7.1: Rouché's theorem

Consider the polynomial $p(z) = z^4 - 5z + 1$. We will show it has exactly one root a with $|a| < 1/4$ and three remaining roots in the annulus $3/2 < |z| < 15/8$.

Restrict $|z| < 1/4$. Then for $p(z) = z^4 - 5z + 1$, we can consider the perturbation to be z^4, for

$$|z^4| = \frac{1}{4^4} \qquad |-5z+1|^2 = 25|z|^2 - 5\operatorname{Re} z + 1$$

Thus z^4's modulus is smaller than $-5z + 1$ in the relevant domain. The unperturbed part has one root $5z + 1 = 0 \implies z = -1/5$. By Rouché's theorem the number of zeros is unchanged and we've shown there is exactly one root with $|a| < 1/4$.

Take the unperturbed part to be $z^4 - 5z$ and the perturbation to be 1. $z^4 - 5z = 0 \implies z^3 - 5 = 0$ and thus the unperturbed part has three roots. Now check that indeed the perturbation is smaller than the

unperturbed part in the desired domain:

$$x^4 - 5x = 1 \Longrightarrow x = \frac{5 \pm \sqrt{29}}{2} \qquad \text{(can take real } x = |z|\text{)}$$

such that

$$\left|\frac{5-\sqrt{29}}{2}\right| < \frac{3}{2} < \frac{15}{8} < \left|\frac{5+\sqrt{29}}{2}\right|$$

- **Open mapping theorem:** $f(z)$ analytic on domain D, $f(z)$ not constant. Then $f(z)$ maps open sets to open sets.

8 Schwarz Lemma

- **Schwarz lemma:** $f(z)$ analytic for $|z| < 1$, $|f(z)| \leq 1$ for $|z| < 1$, and $f(0) = 0$. Then

$$\begin{cases} |f(z)| \leq |z|, & |z| < 1 \\ |f'(0)| \leq 1 \end{cases} \qquad \text{and} \qquad \text{if equality at some point, } f(z) = \lambda z \quad |\lambda| = 1$$

  where λ is a constant.

- **Conformal mapping:**
  - Angle between curves intersecting at z_0 is the angle between their tangent vectors at that point. Tangent vector of γ given to a function is

$$(f \circ \gamma)'(0) = f'(z_0)\gamma'(0) = (f \circ \gamma)'(0) = f'(\gamma(0)) \cdot \gamma'(0)$$

    $t = 0$ is input for which $\gamma(t)$ terminates, $z_0 = \gamma(0)$.
  - Function conformal at z_0 if $(f \circ \gamma_1)(z_0)$ and $(f \circ \gamma_2)(z_0)$ intersect at the same angle as $\gamma_1(z_0)$ and $\gamma_2(z_0)$. (Need to preserve orientation as well!)
  - Conformal mapping of a domain D onto another V is continuous, differentiable, one-to-one, and conformal for all points in D
  - If $f'(z)$ analytic at z_0 and $f'(z_0) \neq 0$, then $f(z)$ is conformal at z_0.

- **Conformal self-mappings of unit disk:** analytic, bijective function from $\mathbb{D} = \{|z| < 1\}$ to itself.
  - $g(z)$ conformal map of the unit disk. $g(0) = 0$. Then $g(z)$ is a rotation: $g(z) = e^{i\phi}z$, $\phi \in [0, 2\pi]$.
  - Conformal self-maps of open disk are (Möbius transformations) of the form

$$f(z) = e^{i\phi}\frac{z-a}{1-\bar{a}z} \qquad |z| < 1$$

- **Pick's lemma:** $f(z)$ analytic, $|f(z)| < 1$ for $|z| < 1$, then

$$|f'(z)| \leq \frac{1-|f(z)|^2}{1-|z|^2}, \qquad |z| < 1$$

  If $f(z)$ conformal self map of unit disk, equality holds. Otherwise strict inequality.

- **Hyperbolic geometry:**
  - Length of γ in hyperbolic metric:

$$L(\gamma) = 2\int_\gamma \frac{|\,\mathrm{d}z\,|}{1-|z|^2}$$

- From Pick's lemma, if $w = f(z)$ conformal self-map of disk

$$\frac{|\,\mathrm{d}w\,|}{1-|w|^2} = \frac{|\,\mathrm{d}z\,|}{1-|z|^2}$$

  So hyperbolic length preserved by conformal self-map of disk.
- **Hyperbolic geodesics:** Two distinct points in $\mathbb{D}$. There is a unique shortest curve (in the hyperbolic metric) passing through both points. It is orthogonal to the unit disk.

- **Spherical geometry:** spherical length is

$$L(\gamma) = 2\int_\gamma \frac{|\,\mathrm{d}z\,|}{1+|z|^2} = 2\int \frac{|\gamma'(t)|}{1+|\gamma(t)|}\,\mathrm{d}t$$